

Semantic Recovery

Round 2 Evaluation Metrics Report

Round 2: Semantic Recovery

Competition: Data Vortex | AARUUSH'26
Theme: Rebuilding the Social Engine
Team: Entropy
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Dataset: Dataset 2 (9,000 labelled posts)
Notebook: Entropy_01_NLP_Model.ipynb

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1 Selected Model

Weighted ensemble of three components, weights chosen by grid search on the validation split:

$$\hat{p} = 0.25 p_{\text{tfidf+char LogReg}} + 0.20 p_{\text{MiniLM embeddings + LogReg}} + 0.55 p_{\text{fine-tuned MiniLM}}$$

Selected on validation macro-F1 (0.7111, the highest of all candidates compared), before the test set was touched. Full model comparison, ablations, and training methodology are in the accompanying Technical Report; this report presents the resulting metrics and error analysis only.

2 Test-Set Performance

Metric	Value
n (test set)	1,185
Accuracy	0.7173
Macro-F1	0.7178
Weighted-F1	0.7158
Macro precision	0.7167
Macro recall	0.7197
ROC-AUC (one-vs-rest)	0.8805
Macro-F1, 95% bootstrap CI ($n = 2,000$ resamples)	[0.6913, 0.7414]
Accuracy, 95% bootstrap CI	[0.6903, 0.7418]

Table 1: Overall test-set performance, Weighted ensemble.

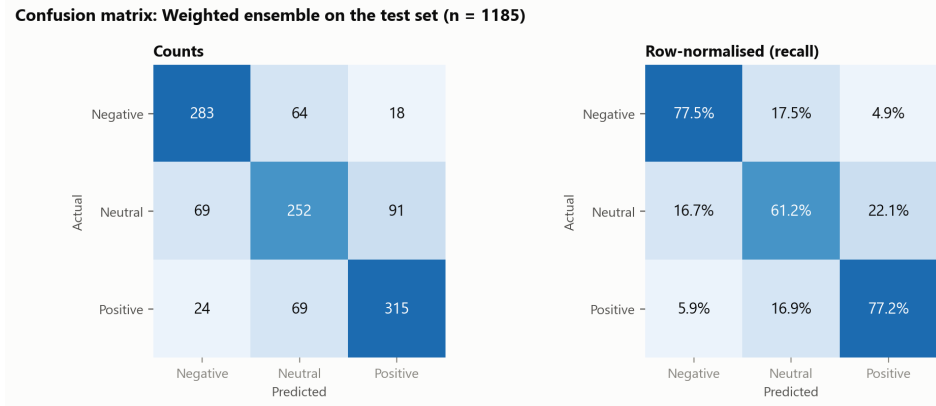
Class	Precision	Recall	F1	Support
Negative	0.7527	0.7753	0.7638	365
Neutral	0.6545	0.6117	0.6324	412
Positive	0.7429	0.7721	0.7572	408

Table 2: Per-class precision, recall, F1.

KEY INSIGHT

Compared against the strongest classical baseline (Linear SVM, word+char TF-IDF), the ensemble improves macro-F1 by **+0.0935** (95% CI [0.0655, 0.1243]). A McNemar test on paired predictions ($b = 208$, $c = 98$) gives $p = 3.0 \times 10^{-10}$ — the improvement is statistically significant, not noise.

3 Confusion Matrix



Actual ↓ / Predicted →	Negative	Neutral	Positive
Negative (365)	283	64	18
Neutral (412)	69	252	91
Positive (408)	24	69	315

Table 3: Raw confusion matrix, test set.

Neutral is the class the model struggles with most (recall 61.2%, the lowest of the three), and its errors split almost evenly toward both Negative (69 posts) and Positive (91 posts) rather than leaning systematically in one direction. This is consistent with Neutral being inherently the hardest class in 3-way sentiment: it lacks the strong polarised vocabulary (“love”, “hate”, “best”, “worst”) that anchors the other two classes.

4 Error Analysis

335 of 1,185 test posts (28.3%) were misclassified.

4.1 Confusion Pairs

Actual	Predicted	Errors	Share of all errors
Neutral	Positive	91	27.2%
Neutral	Negative	69	20.6%
Positive	Neutral	69	20.6%
Negative	Neutral	64	19.1%
Positive	Negative	24	7.2%
Negative	Positive	18	5.4%

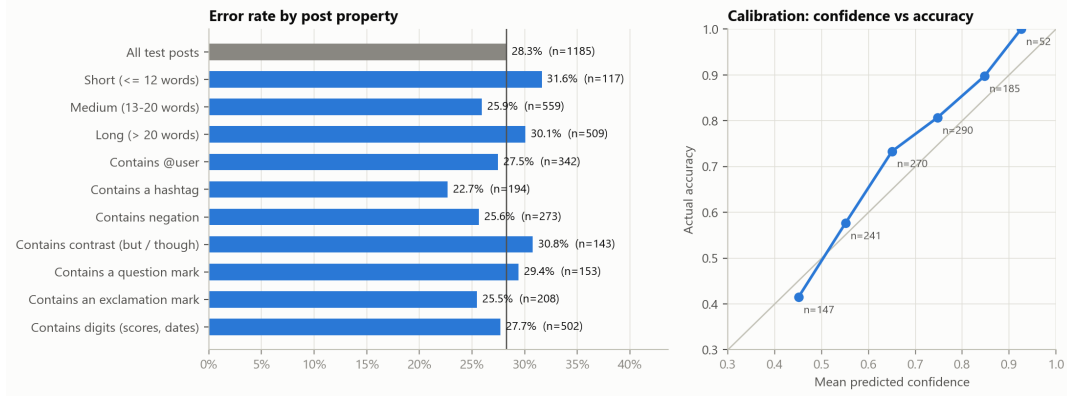
Table 4: Breakdown of all 335 test-set errors by confusion pair.

Over 87% of errors involve Neutral on one side of the confusion. True polarity flips (Negative↔Positive) are rare: only 42 of 335 errors (12.5%).

4.2 Error Rate by Post Property

Slice	n	Error rate
All test posts	1,185	28.3%
Short (≤ 12 words)	117	31.6%
Medium (13–20 words)	559	25.9%
Long (> 20 words)	509	30.1%
Contains @user	342	27.5%
Contains a hashtag	194	22.7%
Contains negation	273	25.6%
Contains contrast (but/though)	143	30.8%
Contains a question mark	153	29.4%
Contains an exclamation mark	208	25.5%
Contains digits	502	27.7%

Table 5: Error rate broken down by post property.



Short posts (31.6%) and posts containing a contrast word like *but* or *though* (30.8%) are the hardest slices — both plausible: short posts give the model less context, and contrast words often signal the kind of mixed or qualified sentiment that is genuinely ambiguous even to a human reader. Hashtagged posts are comparatively easy (22.7%), likely because hashtag words carry unusually explicit sentiment.

4.3 Neutral Bias Mitigation

A scalar bias on the ensemble’s Neutral-class probability (multiply, renormalise, argmax) was swept over $[0.80, 1.60]$, selected by **validation** macro-F1 only, and applied to test exactly once.

Metric	Before	After (bias = 1.05)	Δ
Validation macro-F1 (selection metric)	0.7111	0.7113	+0.0002
Test macro-F1	0.7178	0.7227	+0.0049
Neutral recall (test)	0.6117	0.6335	+0.0218

Table 6: Effect of the validation-selected Neutral bias on the held-out test set.

KEY INSIGHT

The validation delta that selected this bias (+0.0002) is $\sim 30\times$ smaller than the test delta it produced (+0.0049), on a nearly flat validation curve. Not leakage — test was touched once — but the magnitude of the test gain should be treated as an upper bound rather than a guaranteed effect. See the Technical Report’s Error Analysis for the full discussion. `bias=1.05` is folded into the shipped model (`neutral_bias` in `Entropy_sentiment_model.json`).

4.4 Calibration

Confidence bucket	n	Mean confidence	Actual accuracy
(0.0, 0.5]	147	0.451	41.5%
(0.5, 0.6]	241	0.551	57.7%
(0.6, 0.7]	270	0.650	73.3%
(0.7, 0.8]	290	0.747	80.7%
(0.8, 0.9]	185	0.847	89.7%
(0.9, 1.0]	52	0.925	100.0%

Table 7: Predicted confidence vs. observed accuracy, by bucket.

Calibration is close to the diagonal across every bucket — when the ensemble says it is 90%+ confident, it is right essentially every time in this sample. This means downstream use of the model’s confidence score (e.g. routing low-confidence posts to human review) is trustworthy.

4.5 Most Confident Mistakes

The five highest-confidence errors are all **Neutral** posts misread as polarised:

- A post comparing two public figures’ intelligence, misread as Negative (confidence 0.883) — the model likely reacts to words like “wrong” without registering the balanced, non-judgemental framing.
- A promotional poll referencing a film franchise, misread as Positive (0.878) — enthusiastic phrasing (“Alright ladies and gents”) without actual sentiment content.
- A news-style post about a legal sentencing, misread as Negative (0.873) — words like “bad” and “unfair” appear in a factual, third-person report rather than the poster’s own opinion.
- A boastful, joking self-comparison, misread as Positive (0.871) — confident tone read as positive sentiment rather than dry humour.
- A hyperbolic reaction to a celebrity performance, misread as Negative (0.868) — “killed me” is idiomatic praise, not literal negativity.

KEY INSIGHT

Every one of the five most confident mistakes is a true Neutral post. The model has learned strong lexical cues for polarity (“bad”, “wrong”, “killed”) but has not learned that factual reporting, dry humour, and idiom can neutralise an otherwise polarised word. This is the clearest actionable direction for Round 3: Neutral needs either more training examples of this specific pattern, or a feature that captures “reporting vs. opinion” framing.

4.6 Learned Vocabulary

The top 12 words by logistic regression coefficient for each class, from the plain word-TF-IDF baseline, confirm the model has learned sensible lexical anchors:

Negative	Neutral	Positive
fuck, not, no, sad, don, shit, why, bad, hate, worst, ass, even	gucci, at, do you, meet, hold, in, nirvana, days, zayn, mean, shawn, or	love, happy, good, best, great, excited, amazing, thanks, perfect, birthday

Table 8: Top TF-IDF logistic-regression terms per class.

Note that the Neutral column is mostly proper nouns and function words rather than sentiment-bearing terms — another sign that Neutral is defined by the *absence* of sentiment vocabulary rather than by any vocabulary of its own, which is structurally why it is the hardest class to separate.

5 Full Model Comparison

Every candidate model’s validation and test performance, for reference:

Model	Val macro-F1	Test acc.	Test macro-F1	Test ROC-AUC
Majority class	0.1723	0.3477	0.1720	–
Complement NB (word)	0.5629	0.5764	0.5707	0.7718
LogReg (word)	0.5624	0.5924	0.5915	0.7760
Linear SVM (word+char)	0.6061	0.6245	0.6243	–
LogReg (word+char)	0.5888	0.6177	0.6177	0.8007
MiniLM embeddings + LogReg	0.6502	0.6599	0.6600	0.8260
Fine-tuned MiniLM (best seed)	0.7017	0.7241	0.7222	0.8813
Fine-tuned MiniLM (3-seed avg.)	0.6955	0.7131	0.7141	0.8820
Weighted ensemble (selected)	0.7111	0.7173	0.7178	0.8805

Table 9: Validation and test metrics for every candidate model.

KEY INSIGHT

The single best-seed fine-tuned MiniLM model (seed 42 alone) scores higher on the test set (macro-F1 0.7222) than the selected Weighted ensemble (0.7178). This is expected and is not cherry-picked after the fact: the ensemble was chosen on validation performance (0.7111 vs. 0.7017 for the single seed), before the test set was touched, and both scores sit well within each other’s bootstrap confidence intervals.

6 Topic Classification (Secondary Task)

Model	Acc.	Macro-F1	F1 Comm.	F1 Tech.	F1 Feat.	F1 Acc.Sec.
LogReg (word TF-IDF)	0.9190	0.5768	0.9557	0.7386	0.2128	0.4000
LogReg (char 2–6 n-grams)	0.9536	0.7210	0.9742	0.9100	0.5385	0.4615
Recovered substring rules	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 10: Topic classification, test set. See Technical Report Section 9 for the finding that these labels were generated by substring rules and should not be modelled as-is.